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CIVE202

Project #5 Technical Report

Introduction:

This project is based on a dataset from a sample of the Third Generation Simulation (TGSIM) dataset, namely the Foggy Bottom sample. It consists of 200 seconds of trajectory data collected at the intersection of Foggy Bottom in Washington D.C. The sampling interval for collecting this data was 10 Hz (0.1 seconds), giving us 71,361 data points for 564 distinct agents using eight different transportation modes.
This dataset was provided to us in an Excel file (Transformed_TGSIM_Foggy_Bottom_200sec.xlsx). It includes all kinds of vehicles such as autonomous vehicles, regular cars, pedestrians, cyclists, scooters, motorcycles, buses, and trucks.

Main Project Goals:

- The key goals of this project involve quantifying the differences in the actions of AVs, manually controlled passenger vehicles, and pedestrians through
- Analysis of acceleration and deceleration characteristics within intersections and pedestrian zones
- Assessment of speed characteristics within selected lanes

Such analysis will allow us to determine the compatibility of AVs with other road users and their smoothness of operation.

Methods:

The data set was first loaded from a CSV file and cleaned by ensuring the columns were standardized and sorted based on agent IDs and times to maintain consistency.

Agent types were then defined by mapping numerical codes into descriptive labels:

- Automated vehicles (AVs)
- Non-AVs
- Pedestrians

  This dataset was filtered to include these groups only for a more focused comparison. Missing/invalid values were removed to ensure numerical stability.

Spatial Filtering Using Intersection Polygons

For considering only localized interactions, intersection areas were created using polygon boundaries as provided in the example notebook (Polygons.ipynb).

The polygons are the geometrical representation of the intersection area and were used to check if the agent is present within the intersection. For each data point:

- Agent's (x,y) positions
- Observations within the polygon were flagged as "in-intersection"
- A point-in-polygon check was performed

Filtering the dataset this way allows for a more targeted analysis of agent behavior, specifically in high interaction zones.

Derived Motion Metrics

To quantify behavior, several motion-related variables were computed:

- Speed
- Acceleration magnitude

The speed of the object was found based on the values of its velocities in the x and y directions, whereas the acceleration was found on the basis of the accelerations in both x and y directions. The data was sorted on the basis of time and agent ID.

Behavioral Analysis

The analysis focused on comparing automated vehicles (AVs), non-automated passenger vehicles, and pedestrians within localized roadway regions. Since the dataset only covers 200 seconds, the study emphasized short-term interaction patterns rather than complete trajectories. This analysis included:

- Average speeds in the intersection.
- Acceleration and deceleration behavior.
- Frequency of deceleration events
- Lane-based speed consistency across agent types

Visualization

The visualizations used include trajectory graphs, graphs showing the speed distributions, as well as lane-wise comparisons of speeds. The intersections of polygons have been highlighted within the spatial graphs for the areas with high interaction zones.

Results & Discussion:

3.1 Mean Speed and Acceleration by Agent Group

Various visualizations created with the python code were utilized in the analysis process, these were created from the 200 seconds of data that was grouped into three main groups (AV, Non-AV Passenger Vehicle, and Pedestrian). The data for these groups was then summarized into various statistics like mean speed to be utilized for other visualizations and analysis.

Table 1: Speed & Acceleration

	report_group	observations	unique_agents	mean_speed	median_speed	std_speed	mean_signed_accel	median_signed_accel	std_signed_accel	mean_accel_magnitude	median_accel_magnitude
0	AV	1975	1	2.500508	3.000000	2.176278	0.010132	0.0	1.667596	0.328425	0.0
1	Non-AV Passenger Vehicle	47143	157	3.245017	0.000000	4.840877	0.024912	0.0	2.294936	0.534107	0.0
2	Pedestrian	18290	380	1.295091	1.414214	0.832749	0.024866	0.0	1.424222	0.191794	0.0

The mean speed by group through the intersection was created into a bar chart and it can be seen that Non-AV vehicles go through the intersection quicker than the single AV, both of which are much faster than the pedestrians. Additionally, the AV has a significantly lower mean signed acceleration being only 40% of the Non-AV. The distribution of speed is visualized in a box and whisker chart for the three groups. All the groups have speeds skewed towards 0 m/s due to stopping, but looking upwards the max speed of the AV is only slightly larger than the Inner Quartile Range (IQR) of the Non-AV. The range of speeds vary a lot being ~7m/s for the AV and ~15m/s for the Non-AV not including the outliers. The Non-AV plot also has several outliers at speeds double the max AV speed.

Figure 1: Average Speed Bar Chart

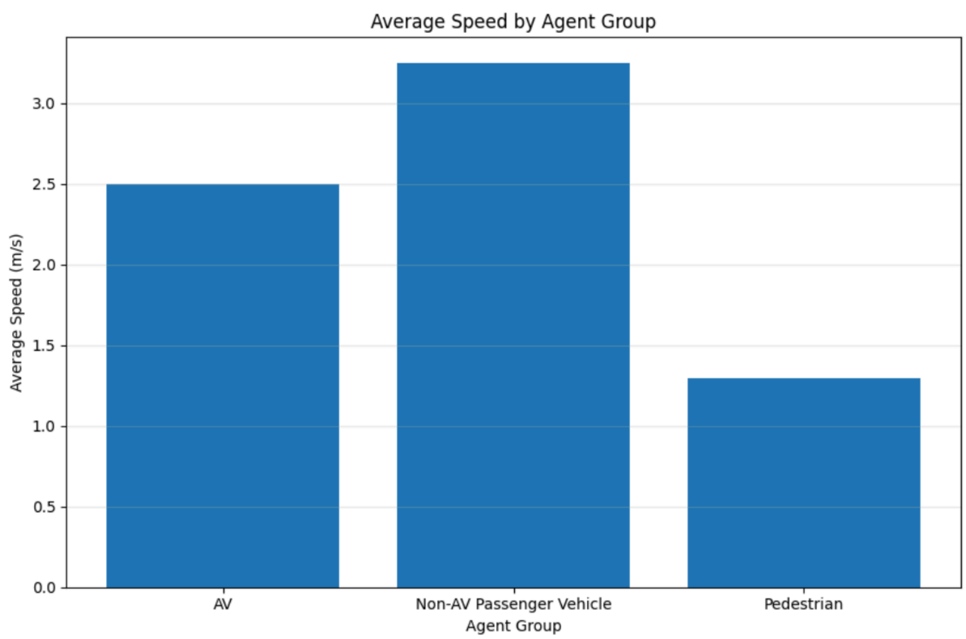


Figure 2: Average Acceleration Bar Chart

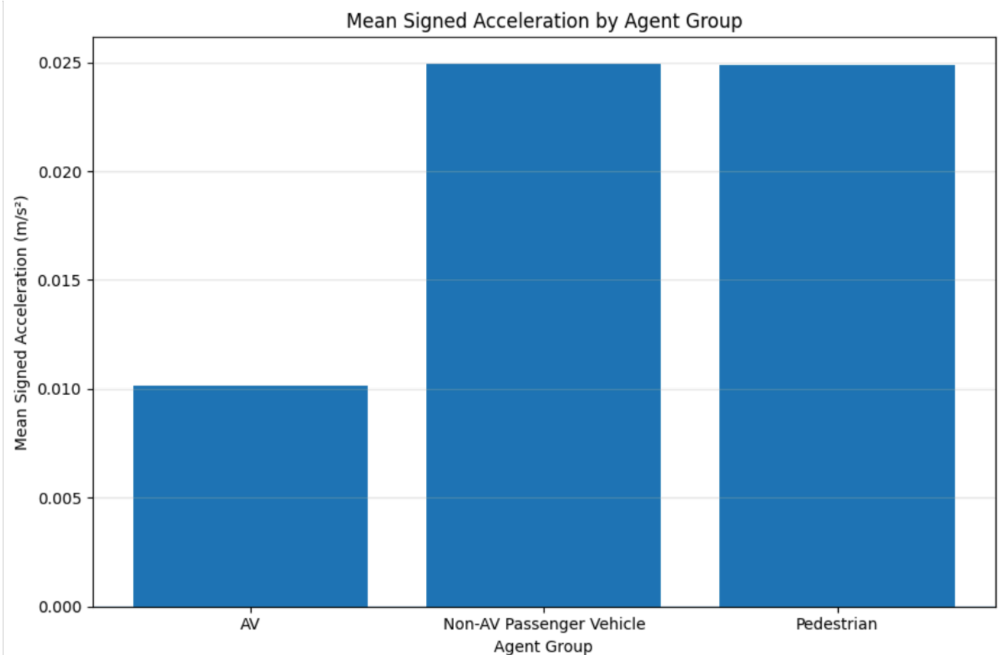
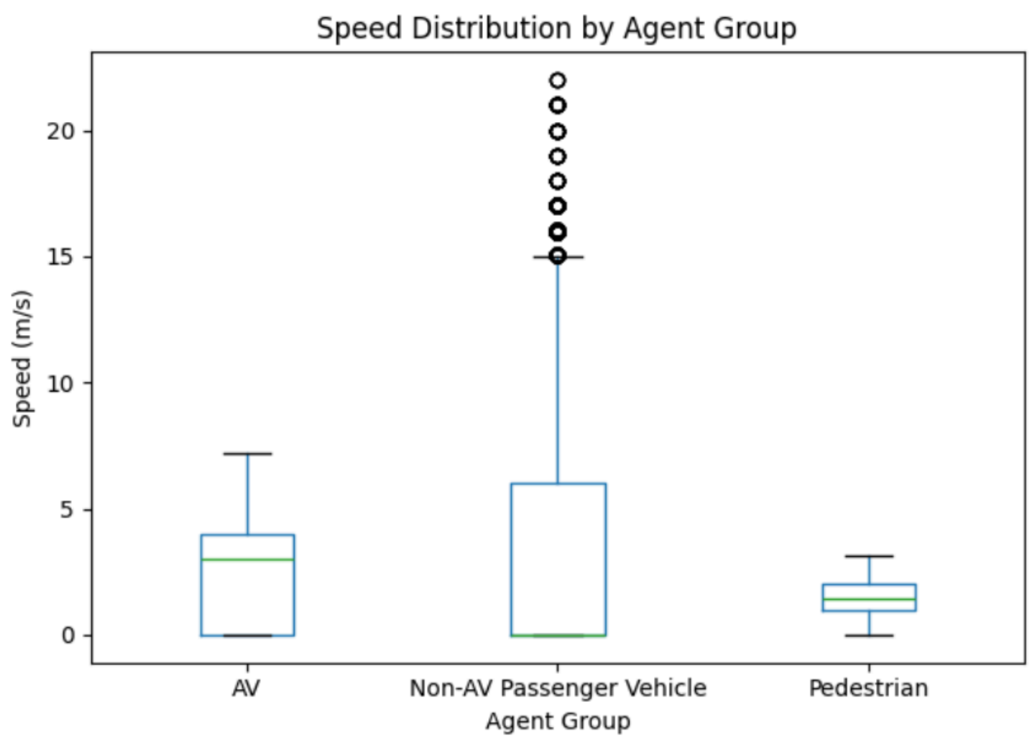


Figure 3: Speed Distribution Visualization



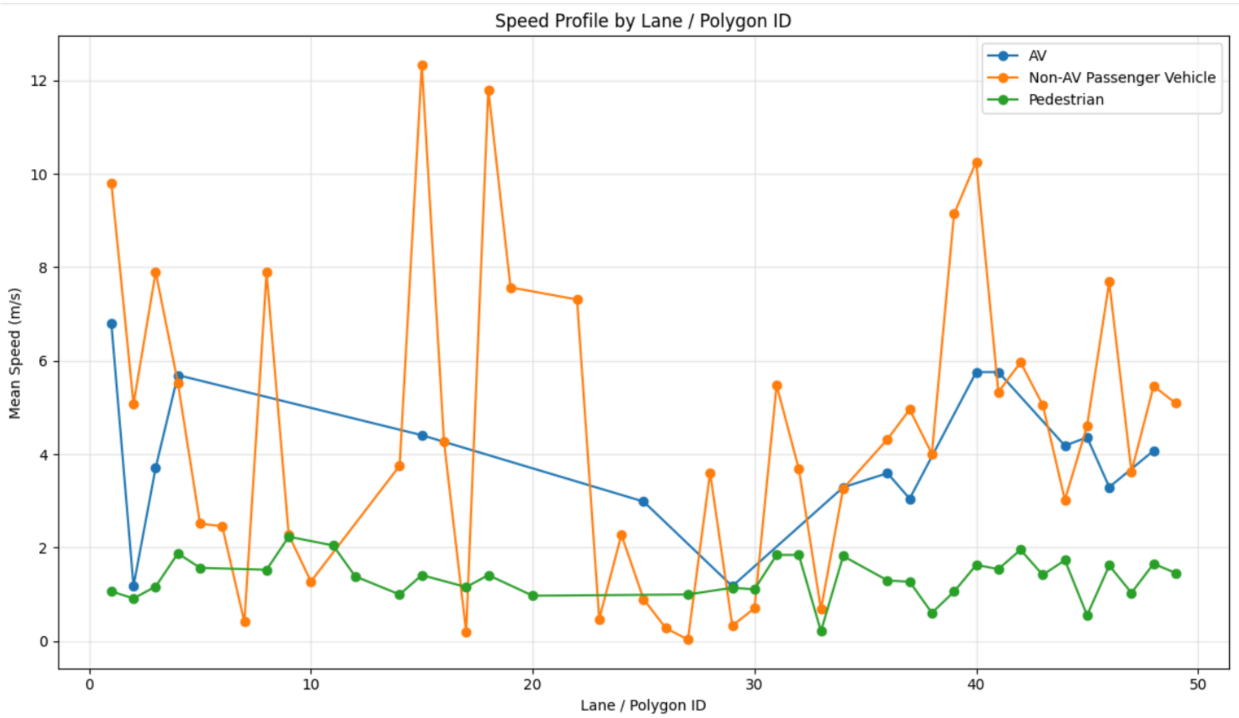
Relevance: The bar charts of mean speed and signed acceleration by agent groups reveal that the AV goes through the intersection with more control as it has a lower speed and doesn't spend as much time accelerating to greater speeds. This is supported by the speed distribution plot too, The Non-AV vehicles have speeds that span a range around double that of the AV showing a lack of consistency.

3.2 Speed Profile by Lane

The intersection data consists of different sections as lane / polygon IDs, looking at data by grouping it into the three main agent groups and by ID to look at how the agents travel through the intersection. A visualization of this was created as a scatter plot with a series of data for each group. By looking at the speeds for the different IDs it can be seen that between Lane 23-30 is a common spot for the agents to be slowing down, idling, or starting to accelerate which shows that that is where the intersection itself is and traffic lights control the traffic, this is also the case for lanes 5-11. The lanes around these are the sections of the intersection where people are approaching or exiting the intersection.

Shifting the analysis to the agent groups - pedestrians are, as expected, the slowest throughout the intersection without much fluctuation. Now comparing the AV vs the Non-AVs, there is significant fluctuation in the Non-AV data compared to the AV data. The AV data, as a nature of being composed of a single unique agent is going to have less fluctuation; despite this both series follow similar trends as they go through the intersection with the Non-AVs speed spiking much higher.

Figure 4: Lane ID Speed Profile Plot



Relevance: This plot further reinforces the idea that the AV goes through the intersection with more control and consistency as can be seen with the lack of major fluctuation present in the Non-AV vehicles' data line.

3.3 Hard Deceleration Events

Hard deceleration events, as defined in the python code, are when the signed acceleration of the agent exceeds the threshold of -2.0. Python was then used to find the number of hard deceleration events per agent group which was then used to find statistics regarding it to be put in a table. Total hard deceleration events cannot solely be used for analysis because of the varying number of unique agents for the groups. Looking at Table 2 it can be seen that mean hard deceleration events is much higher for the single AV than the Non-AV vehicles. Pedestrians with the nature of them being on the sidewalks or crosswalks encounter fewer situations that require hard negative acceleration on the average trip through the intersection.

Table 2: Deceleration Events

	report_group	agents_in_group	total_hard_decel_events	mean_hard_decel_events_per_agent	median_hard_decel_events_per_agent
0	AV	1	31	31.000000	31.0
1	Non-AV Passenger Vehicle	157	1230	7.834395	7.0
2	Pedestrian	380	217	0.571053	0.0

The mean hard deceleration events per agent type were plotted as a bar chart visualization, looking at this it's even easier to see the stark contrast between agent types. Furthermore, when looking at the AV acceleration visualization, it can be seen that almost every single time the AV braked it crossed the threshold.

Figure 5: Hard Deceleration Events Bar Chart

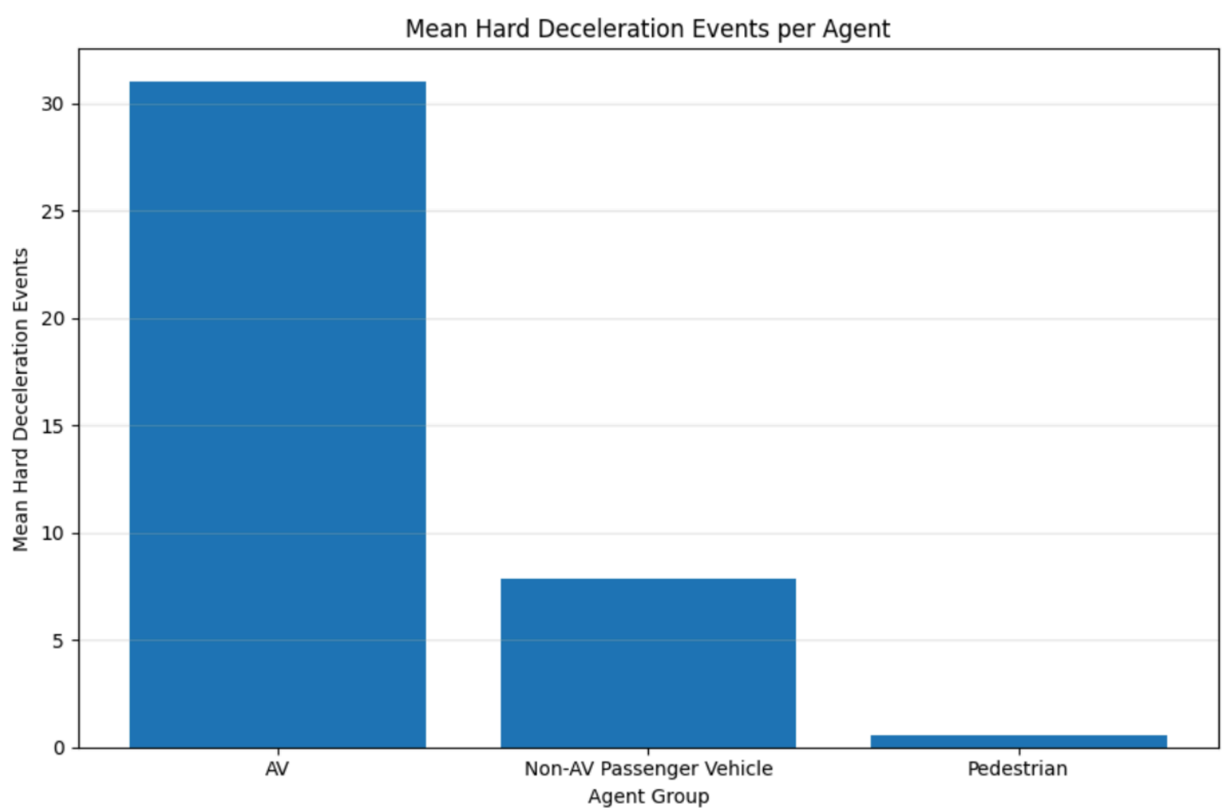
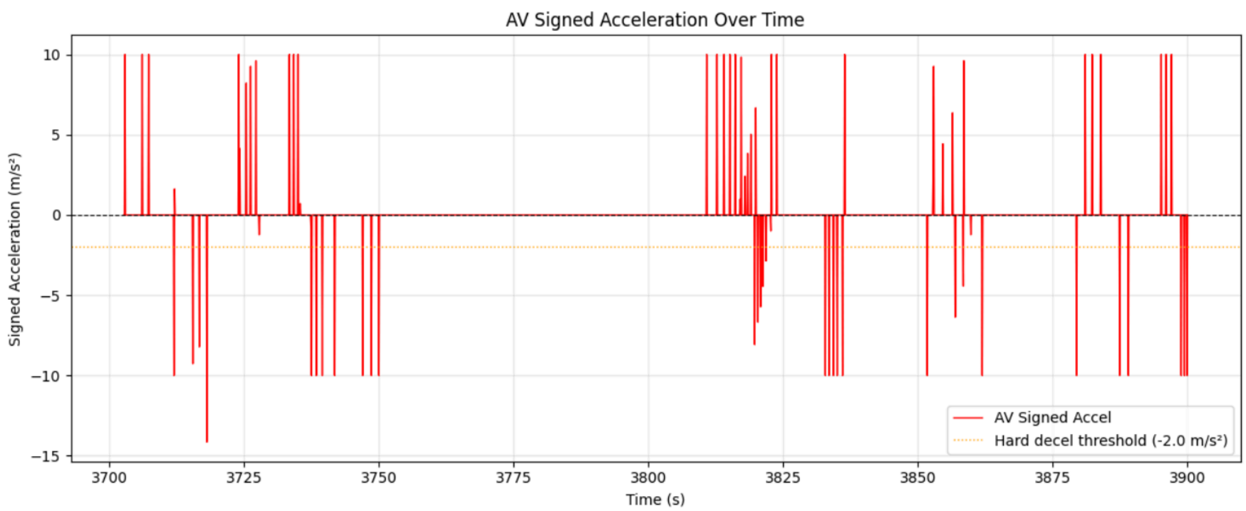


Figure 6: AV Acceleration Visualization



Relevance: The bar chart and the statistics table both contrast the conducted analysis previously made, when considering hard deceleration events it is the Non-AV vehicles that seem to have more control and travel smoother due to needing to 'slam' on the brakes less than 25% of the time as the AV and/or brakes much smoother due to visual foresight.

3.4 Agent Travel Through The Intersection

From the basic analysis done on the data and visualizations conclusions can be made about the agents group when looking at speed the AV looks generally more controlled and smooth through the intersection where the contrary is suggested with hard breaking events, but different parts of it all need to come together.

From the engineering team's analysis, the AV behaves smoother, more controlled, and safer with caveats. Figure 3 especially demonstrates the difference in the tendencies of the agent types with the AV being much more controlled within the intersections and pedestrian zones. Speed is a very significant factor in drawing this conclusion because one's speed and consistency influence how others around will be able to react to the vehicle. By having a lower range of speed and mean speed the AV is generally safer. Notably part of the difference in speed behavior comes from being an AV vs human. Human drivers like to go as fast as they reasonably can (generally) and therefore push their speed and accelerate more than any AV programmed to stay within the legal speed limit ever would.

During analysis of the agents the AV's large amount of hard deceleration events brings up questions regarding it. Quick changes in speed (acceleration) can throw other vehicles off as its less predictable or avoidable, however, the quantity of events comes from the fact that the AV 'is' an AV. human drivers and pedestrians use their vision and reasoning to build anticipation. An AV being coded to work a certain way and equipped with certain things has much more limited foresight. Because of this an AV is built to brake harder at the prospect of necessity, think a pedestrian hesitating to cross the street. A human driver could start to brake and then very quickly see them stopping their crossing which in response the driver speeds back up. Conversely, an AV is going to see the pedestrian starting to cross its path and try very hard to avoid that without awareness the human has.

The engineering team points at lots of potential for larger adoption of AVs to lead to safer roads due to their control and consideration. However they also suggest that further development and interaction analysis is required in order to ensure something with so much power and mass can be absolutely credited as the safest option.